

# 1 Strategic Reports from Teams

Teams report private value; demand and emissions are observable

| Team                          | Demand $d_i$ (h)          | Reported $r_i$ | CO <sub>2</sub> e $e_i$ (kg) | Score $s_i = r_i - \lambda e_i$ |
|-------------------------------|---------------------------|----------------|------------------------------|---------------------------------|
| <div><div></div> Team A</div> | <div><div></div> 25</div> | 9              | 6.0                          | 6.0                             |
| <div><div></div> Team B</div> | <div><div></div> 20</div> | 6              | 4.8                          | 3.6                             |
| <div><div></div> Team C</div> | <div><div></div> 15</div> | 3              | 3.6                          | 1.2                             |
| <div><div></div> Team D</div> | <div><div></div> 25</div> | 9              | 6.0                          | 6.0                             |
| <div><div></div> Team E</div> | <div><div></div> 20</div> | 6              | 4.8                          | 3.6                             |
| <div><div></div> Team F</div> | <div><div></div> 15</div> | 3              | 3.6                          | 1.2                             |

## Information structure

- Private: project value  $r_i$  (may be misreported)
- Observable: demand  $d_i$  and estimated emissions  $e_i$
- Carbon weight  $\lambda = 0.5$ ; capacity  $K = 100$  GPU-hours
- Total demand = 120 GPU-hours  $> K$  (contested)

# 2 Carbon-Aware VCG Allocation

Select the feasible set maximizing carbon-adjusted reported value

## Optimization problem

maximize  $\sum_{i \in S} (r_i - \lambda e_i)$

subject to  $\sum_{i \in S} d_i \leq K$

$K = 100$  GPU-hours,  $\lambda = 0.5$

$r_i$  : reported value

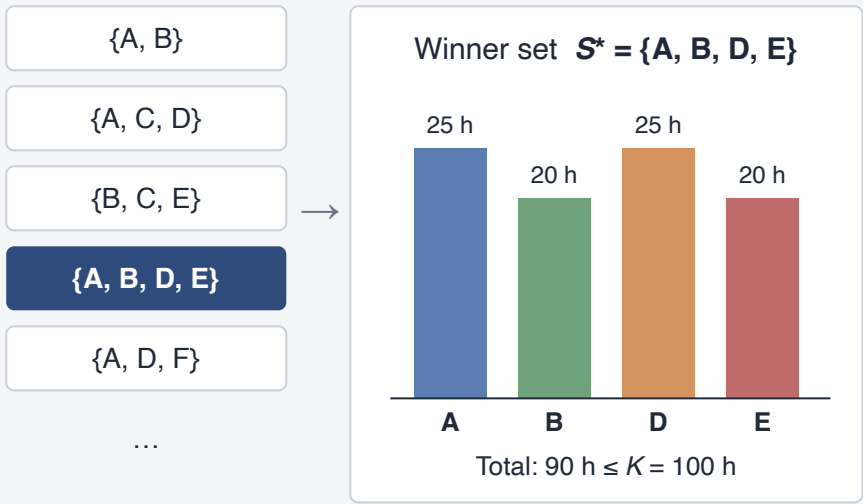
$d_i$  : GPU-hours requested

$e_i$  : estimated CO<sub>2</sub>e (kg)

$\lambda$  : carbon weight

$K$  : capacity

## Exhaustive search over feasible subsets



## VCG payment for each winner $i$

$p_i = W^*_{-i} - \sum_{j \in S^* \setminus \{i\}} s_j$

$p_i$  : VCG payment of team  $i$

$W^*_{-i}$  : optimal value without  $i$

$s_j$  : carbon-adjusted score of  $j$

# 3 Outcomes and Properties

Transparent allocation, payments, and carbon-aware efficiency

## Resource allocation ( $K = 100$ GPU-hours)



## Payments (VCG)

24 priority credits each

|                               |    |
|-------------------------------|----|
| <div><div></div> Team A</div> | 24 |
| <div><div></div> Team B</div> | 24 |
| <div><div></div> Team D</div> | 24 |
| <div><div></div> Team E</div> | 24 |

Total 96 credits

## Outcomes (selected set)

|                                     |      |
|-------------------------------------|------|
| Total value $\sum r_i$              | 30   |
| Carbon-adjusted score               | 19.2 |
| Total CO <sub>2</sub> e (kg)        | 21.6 |
| kg CO <sub>2</sub> e per unit value | 0.72 |

## Comparison with first-come-first-served (FCFS)

|                                       |       |
|---------------------------------------|-------|
| FCFS score, illustrative single order | 15.6  |
| FCFS mean score over all 720 orders   | 16.96 |
| Carbon-aware VCG score                | 19.2  |

✓ No FCFS order exceeds the VCG score

## Key economic properties

|                                       |                             |                              |
|---------------------------------------|-----------------------------|------------------------------|
| Truthful reporting is dominant (DSIC) | Social welfare maximization | Carbon-aware and transparent |
|---------------------------------------|-----------------------------|------------------------------|

Incentive-compatible  
truthful reporting (DSIC)

Exact enumeration  
720 FCFS orders / 64 VCG subsets

Lower emissions  
per unit of value

Practical insights  
for university GPU governance